Toeplitz Matrix (easy)

Problem Statement

Given an `m x n` matrix, determine if a given matrix is a `Toeplitz matrix`.

A Toeplitz matrix is one in which every diagonal from `top-left` to `bottom-right` has the same elements.

In other words, the matrix should have the property that each element is equal to the element diagonally down to its right.

Example 1:

- **Input**:

```
[[1,2,3],
 [4,1,2],
 [5,4,1]]
```

- **Expected Output**: true
- **Justification**: All diagonals have the same elements. Diagonals of the above matrix are [5], [4, 4], [1, 1, 1], [2, 2], and [3].

Example 2:

- **Input**:

```
[[1,2],
 [3,4]]
```

- **Expected Output**: false
- **Justification**: The diagonal are [3], [1, 4], and [2]. A diagonal [1, 4] doesn't have the same elements.

Example 3:

- **Input**:

```
[[7,7,7],
 [7,7,7],
 [7,7,7]]
```

- **Expected Output**: true
- **Justification**: Each diagonal, although containing the same repeated element (7), satisfies the condition of a Toeplitz matrix.

Try it yourself

Try solving this question here:

Java

```java
public class Solution {
    // Method to check if a matrix is a Toeplitz matrix
    public boolean isToeplitzMatrix(int[][] matrix) {
        // ToDo: Write Your Code Here.
        return false;
    }
}
```

Test Case | Run Code Result | Stdout

[[1,2,3,4],[5,1,2,3],[9,5,1,2]]

Use Example Testcases | Run | Submit